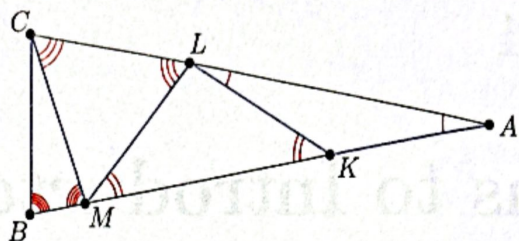


we apply this step one more time to get $\angle CBA = \angle BMC = 3\alpha + \alpha = 4\alpha$. Since triangle ABC itself is isosceles we have

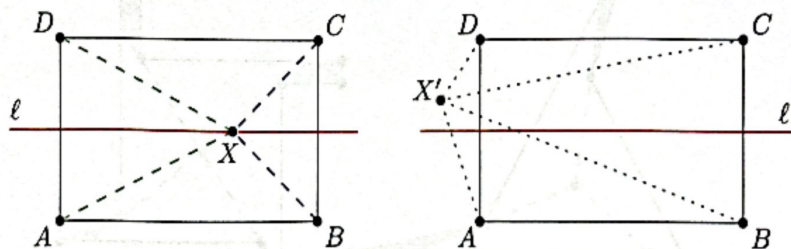
$$180^\circ = \angle ACB + \angle BCA + \angle BAC = 4\alpha + 4\alpha + \alpha = 9\alpha.$$



Thus the answer is $\alpha = 20^\circ$.

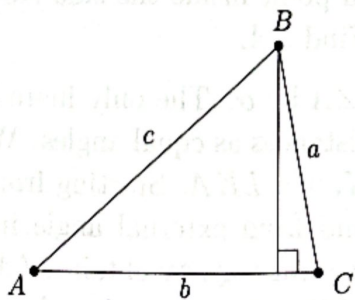
3. Let $ABCD$ be a rectangle. Find the locus of points X such that $AX + BX = CX + DX$.

Solution. Draw the common perpendicular bisector ℓ of BC and AD and assume it is horizontal with AB below ℓ . Points X on ℓ obviously satisfy the desired condition since $BX = CX$ and $AX = DX$. Points X' which are above ℓ have $BX' > CX'$ and $AX' > DX'$, therefore they don't fulfil the condition. For analogous reason we can also exclude points below ℓ . Thus the locus is exactly the line ℓ .



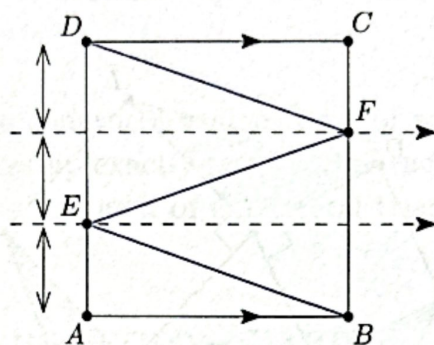
4. Let $a < b < c$ be the sides of a triangle ABC . Prove that $h_b < b$, where h_b is the B -altitude of triangle ABC .

Proof. Since the B -altitude is the shortest distance from B to line AC , we certainly have $h_b \leq a$ and since $a < b$ the result follows.



5. [AIME 2011] On square $ABCD$, point E lies inside the side AD and point F lies inside the side BC , so that $BE = EF = FD = 30$. Find the area of square $ABCD$.

Solution. If we place AB horizontally and draw horizontal lines also through points E and F , we see that we have divided the square into six pairwise congruent triangles (HL).



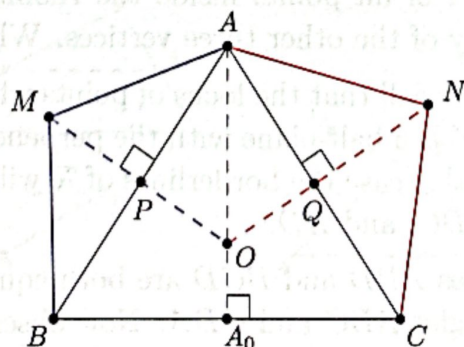
Thus, $AE = \frac{1}{3}a$, where a is the side length of the square. From the Pythagorean Theorem we learn

$$30^2 = BE^2 = a^2 + \left(\frac{1}{3}a\right)^2 = \frac{10}{9}a^2,$$

or equivalently $a^2 = 810$, which is our final answer.

6. Let ABC be a triangle with $AB = AC$. Isosceles triangles ABM and ACN with bases AB and AC are erected outside triangle ABC . Prove that the altitudes (possibly extended) $MP \perp AB$, $NQ \perp AC$ and $AA_0 \perp BC$ are concurrent.

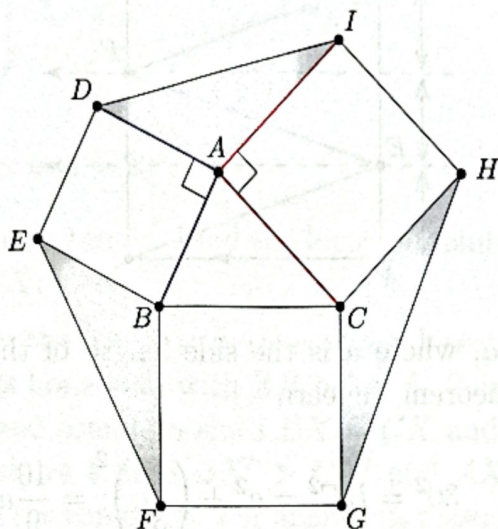
Proof. In isosceles triangle ABM the altitude MP coincides with the perpendicular bisector of AB . Similarly, NQ is the perpendicular bisector of AC and AA_0 is the perpendicular bisector of BC . Since the altitudes are in fact the perpendicular bisectors of triangle ABC , they concur at its circumcenter O .



7. Squares $ABED$, $BCGF$, $CAIH$ are erected externally from the sides of triangle ABC . Show that triangles AID , BEF , and CGH have equal area.

Proof. We turn our attention to triangle DAI . We have $AD = AB$ and $AI = AC$ and also

$$\angle IAD = 360^\circ - 90^\circ - 90^\circ - \angle BAC = 180^\circ - \angle BAC.$$



Thus we can compute the area K_A of the triangle DAI as

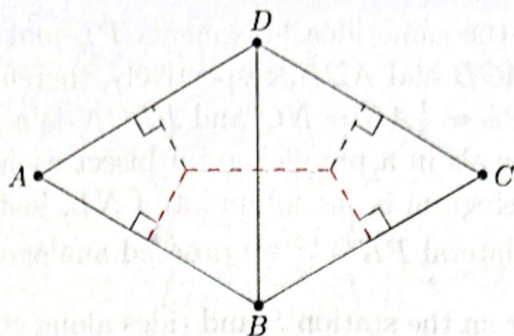
$$\begin{aligned} K_A &= \frac{1}{2} AD \cdot AI \cdot \sin \angle IAD = \frac{1}{2} AB \cdot AC \cdot \sin(180^\circ - \angle BAC) \\ &= \frac{1}{2} AB \cdot AC \cdot \sin \angle BAC \end{aligned}$$

which is exactly the area of triangle ABC ! Thus by symmetry all three triangles have area equal to that of triangle ABC and the conclusion follows.

8. [AMC12 2011] Rhombus $ABCD$ has side length 2 and $\angle B = 120^\circ$. Region $\mathcal{R}$ consists of all points inside the rhombus that are closer to vertex B than any of the other three vertices. What is the area of $\mathcal{R}$?

Solution. First, recall that the locus of points which are closer to point X than to point Y is a half-plane with the perpendicular bisector of XY as borderline. In this case the borderlines of $\mathcal{R}$ will be the perpendicular bisectors of BA , BC , and BD .

Note that triangles ABD and BCD are both equilateral as BD bisects the congruent angles ABC and CDA . Now observe that if we connect midpoint of each side of triangle ABD with its center, we divide the



triangle into three congruent regions, one of which is exactly one half of $\mathcal{R}$. Thus $\mathcal{R}$ takes up exactly one third of each of triangles ABD and BCD . Calculating the area of equilateral triangle we find the desired area K as

$$K = \frac{1}{3}([ABD] + [BCD]) = \frac{2}{3}[ABD] = \frac{2}{3} \cdot \frac{\sqrt{3}}{4} AB^2 = \frac{2}{3}\sqrt{3}.$$

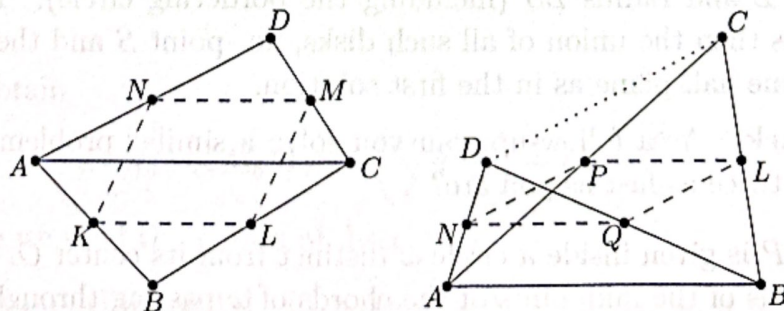
9. Varignon¹ parallelogram

Let $ABCD$ be a quadrilateral and denote by K, L, M, N the midpoints of the sides AB, BC, CD, DA , respectively.

- Prove that $KLMN$ is a parallelogram.
- Let P, Q be the midpoints of the diagonals AC, BD , respectively. Prove that $PLQN$ and $PKQM$ are also parallelograms, moreover with the same center.

Proof.

- The key is to realize that both KL and NM are midlines in some triangles. Namely, in triangles ABC and ADC . Thus $KL \parallel AC \parallel NM$ and also $KL = \frac{1}{2}AC = NM$. This ensures that $KLMN$ is a parallelogram.



¹Pierre Varignon (1654–1722) was a French mathematician.

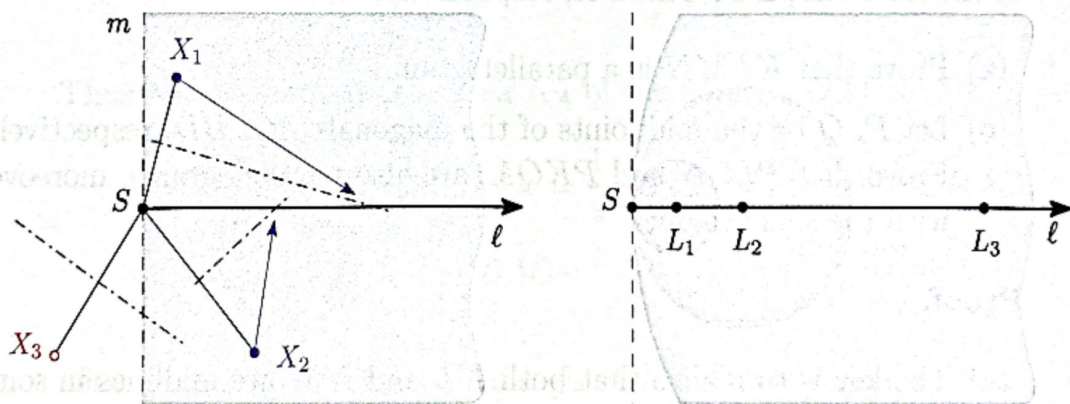
- (b) We apply the same idea. Segments PL and NQ are midlines in triangles ACB and ADB , respectively, therefore again $PL \parallel AB \parallel NQ$ and $PL = \frac{1}{2}AB = NQ$ and $PLQN$ is a parallelogram. Also, since diagonals in a parallelogram bisect each other, the center of this parallelogram is the midpoint of NL , just like in (a).

For quadrilateral $PKQM$ we proceed analogously.

10. A bus departs from the station S and rides along straight (infinite) road ℓ . Determine the locus of points in the plane from which you can catch the bus if you start running at the time of the departure and you are as fast as the bus.

Solution. We can clearly catch the bus from S . For any other point X catching the bus from X is equivalent to finding a point on the ray ℓ (let's assume it's horizontal and points to the right) which is closer to X than to S (or equidistant from them). This happens if and only if the perpendicular bisector of XS intersects ℓ or in other words if and only if the angle between ℓ and XS is less than 90° .

Thus, denoting by m the perpendicular to ℓ through S , the answer is "point S and the half-plane consisting of the points to the right of m ".



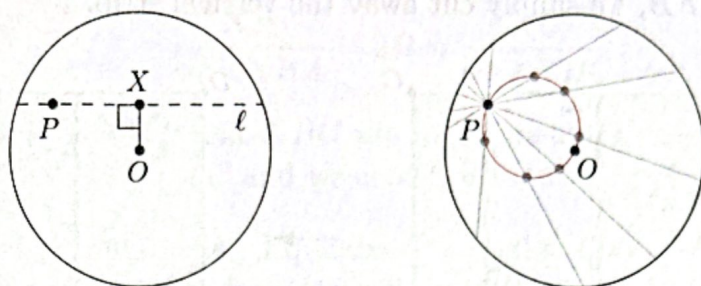
Second Solution. Let L be any point on ℓ . The locus of points, from which the bus can be caught at precisely this point L , is the disk with center L and radius LS (including the bordering circle). The desired locus is then the union of all such disks, i.e. point S and the interior of the same half-plane as in the first solution.

Remark. As a follow-up, can you solve a similar problem where the bus is twice as fast as you are?

11. Point P is given inside a circle ω distinct from its center O . Determine the locus of the midpoints of the chords of ω passing through P .

Solution. First observe that if we draw a chord through both P and O , then its midpoint is O . Now consider some other chord ℓ and denote its

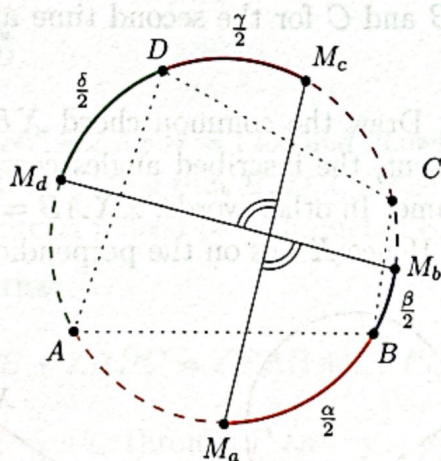
midpoint by X . Clearly, as O is the center of the circle, it is equidistant from the endpoints of ℓ and thus it lies on its perpendicular bisector.



In other words $OX \perp \ell$. If $X \neq P$, this means $\angle OXP = 90^\circ$. Thus, we are restricted to a circle with diameter OP and it is easy to verify that all points of this circle (including P) are indeed midpoints of some chord passing through P .

12. Let $ABCD$ be a quadrilateral inscribed in circle ω and let M_a, M_b, M_c, M_d be the midpoints of the arcs AB, BC, CD, DA not containing points C, D, A , and B , respectively. Prove that $M_a M_c \perp M_b M_d$.

Proof. We divide the circle into the four arcs AB, BC, CD , and DA and label the corresponding inscribed angles as $\alpha, \beta, \gamma, \delta$. Now we calculate the angle between chords $M_a M_c$ and $M_b M_d$ as sum of the corresponding inscribed angles (see Corollary 1.32).



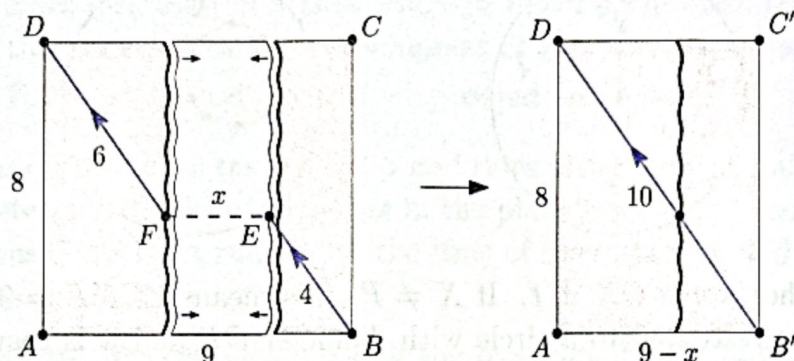
We obtain

$$\angle(M_a M_c, M_b M_d) = \left(\frac{\alpha}{2} + \frac{\beta}{2}\right) + \left(\frac{\gamma}{2} + \frac{\delta}{2}\right) = 90^\circ,$$

where we used the notion of directed angles.

13. [based on AIME 2011] In rectangle $ABCD$, $AB = 9$ and $BC = 8$. Points E and F lie inside rectangle $ABCD$ so that $EF \parallel AB$, $BE \parallel DF$, $BE = 4$, $DF = 6$, and E is closer to BC than F . Find EF .

Solution. We draw our diagram so that AB is horizontal and draw vertical lines through points E and F . In order to connect the segments DF and EB , we simply cut away the vertical strip.



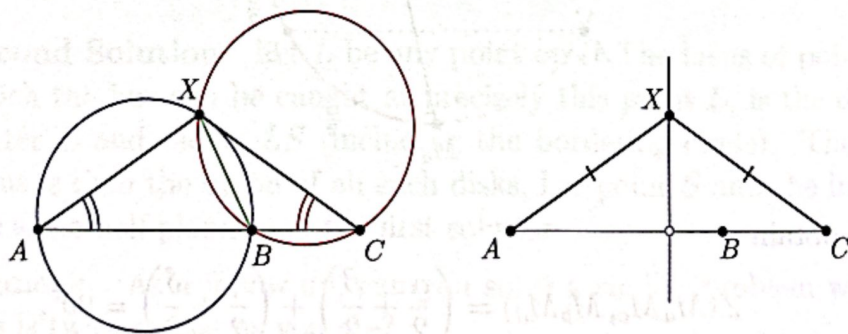
If we denote EF by x , Pythagorean theorem in the newly formed right triangle gives

$$(9 - x)^2 + 8^2 = 10^2 \quad \text{or} \quad (9 - x)^2 = 6^2.$$

Since $x < 9$ (EF lies inside $ABCD$) the solution is $x = 3$.

14. Distinct points A, B, C lie on a line in this order. Circle ω_1 of radius R passing through A and B intersects circle ω_2 of the same radius R and passing through B and C for the second time at X . Find the locus of X as R varies.

First Solution. Draw the common chord XB . Since the circles ω_1 and ω_2 are congruent, the inscribed angles corresponding to the same arc XB are the same. In other words, $\angle XAB = \angle XCB$ so the triangle AXC is isosceles. Hence X lies on the perpendicular bisector of AC .



On the other hand, any such point X' distinct from the midpoint of AC can be attained as the circumcircles of $X'BC$ and $X'AB$ have the same radii. Thus the locus is the perpendicular bisector of AC without the midpoint of AC .

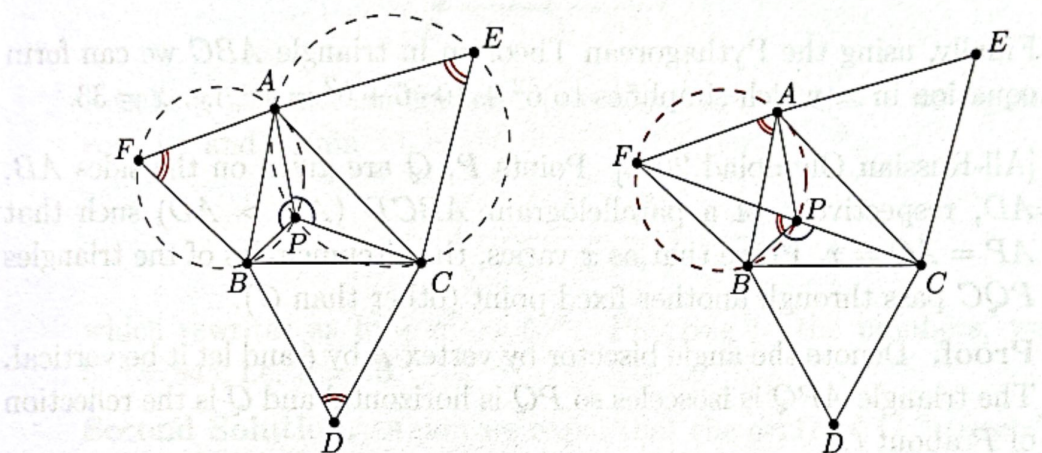
Second Solution. Using the Extended Law of Sines in triangles AXB , BXC we obtain

$$\frac{XA}{\sin \angle XBA} = 2R = \frac{XC}{\sin \angle XBC}.$$

As the angles XBA and XBC are supplementary, their sines are the same. Hence $XA = XC$ and we continue as in the first solution.

15. Let ABC be a triangle. Equilateral triangles BCD , CAE , ABF are erected outwards from its sides. Show that the circumcircles of these equilateral triangles and the lines AD , BE , CF pass through one point.

Proof. Let P be the second intersection of the circumcircles of triangles ABF and ACE .



Then $\angle APB = 180^\circ - \angle AFB = 120^\circ$ and likewise $\angle APC = 120^\circ$ hence also $\angle BPC = 120^\circ$ implying that B, D, C, P lie on a single circle. Thus, the three circumcircles indeed pass through a common point.

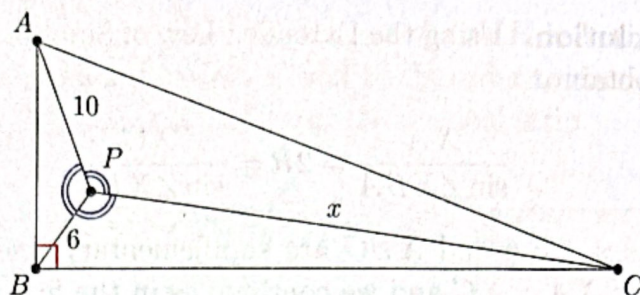
Next we observe that

$$\angle FPC = \angle FPB + \angle BPC = \angle FAB + \angle BPC = 60^\circ + 120^\circ = 180^\circ.$$

Hence the line CF passes through P and by symmetry BE and AD pass through it too. We may conclude.

16. [AIME 1989] Triangle ABC has right angle at B , and contains a point P for which $PA = 10$, $PB = 6$, and $\angle APB = \angle BPC = \angle CPA$. Find PC .

Solution. Denote the length of PC by x . Note that since the angles by P are all equal to 120° , the squares of the side lengths of triangle ABC can be expressed in terms of x by the Law of Cosines applied to triangles ABP , BCP , and CAP .



Keeping in mind that $-2 \cos 120^\circ = 1$, we obtain

$$AB^2 = 10^2 + 6^2 + 10 \cdot 6,$$

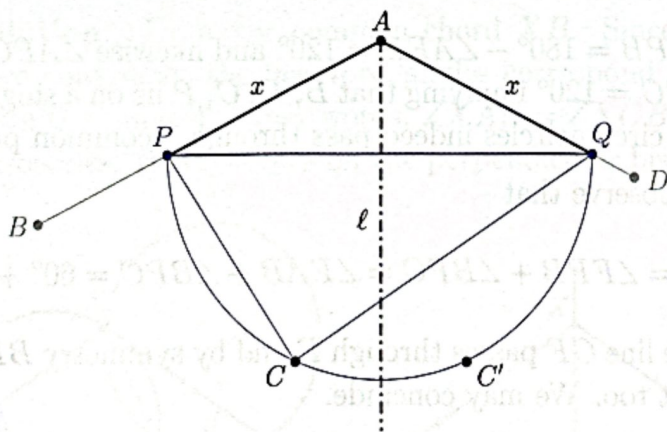
$$BC^2 = 6^2 + x^2 + 6x,$$

$$CA^2 = x^2 + 10^2 + 10x.$$

Finally, using the Pythagorean Theorem in triangle ABC we can form equation in x , which simplifies to $6^2 + 10 \cdot 6 + 6^2 = 4x$, i.e. $x = 33$.

17. [All-Russian Olympiad 2005] Points P , Q are given on the sides AB , AD , respectively, of a parallelogram $ABCD$ ($AB > AD$) such that $AP = AQ = x$. Prove that as x varies, the circumcircles of the triangles PQC pass through another fixed point (other than C).

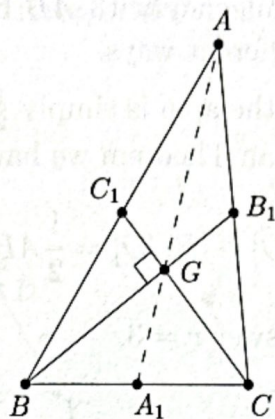
Proof. Denote the angle bisector by vertex A by ℓ and let it be vertical. The triangle APQ is isosceles so PQ is horizontal and Q is the reflection of P about ℓ .



Thus the whole circumcircle of triangle CPQ is symmetric about ℓ and since it passes through C , it also passes through its reflection C' in ℓ , which is the fixed point we were to find (note that $AB > AD$ implies $C' \neq C$).

18. In triangle ABC , medians BB_1 and CC_1 are perpendicular. Given that $AC = 19$ and $AB = 22$, find BC .

First Solution. We draw the third median AA_1 and recall that medians are concurrent at the centroid G . As A_1 is the midpoint of a hypotenuse in right triangle BCG , we have $A_1G = A_1B$ which (since medians “trisection” each other) may be rewritten as $\frac{1}{3}AA_1 = \frac{1}{2}BC$.



Recalling the median formula (see Corollary 1.24(a)), we square the equality and obtain

$$\frac{1}{9} \cdot \left(\frac{b^2 + c^2}{2} - \frac{a^2}{4} \right) = \frac{1}{4}a^2,$$

which rewrites as $b^2 + c^2 = 5a^2$. Plugging in the numbers, we find $5a^2 = 845$, i.e. $a = 13$.

Second Solution. Again we recall that the centroid G “trisects” medians. Denote their lengths by $BB_1 = 3y$, $CC_1 = 3z$.

Pythagorean theorems in right triangles BGC_1 and CGB_1 yield

$$\left(\frac{c}{2}\right)^2 = 4y^2 + z^2 \quad \text{and} \quad \left(\frac{b}{2}\right)^2 = y^2 + 4z^2.$$

Now we could plug in the values of b and c and solve these equations for y and z but observe that we are only interested in

$$BC^2 = BG^2 + CG^2 = 4y^2 + 4z^2.$$

Hence we sum the equations instead. After multiplying by $\frac{4}{5}$ we get $BC^2 = \frac{4}{5}(b^2 + c^2)$ which gives $BC = 13$ again.

Third Solution. We apply the perpendicularity criterion (see Proposition 1.22) for quadrilateral BCB_1C_1 and obtain

$$a^2 + \left(\frac{a}{2}\right)^2 = \left(\frac{b}{2}\right)^2 + \left(\frac{c}{2}\right)^2,$$

from which we again find $BC = 13$.

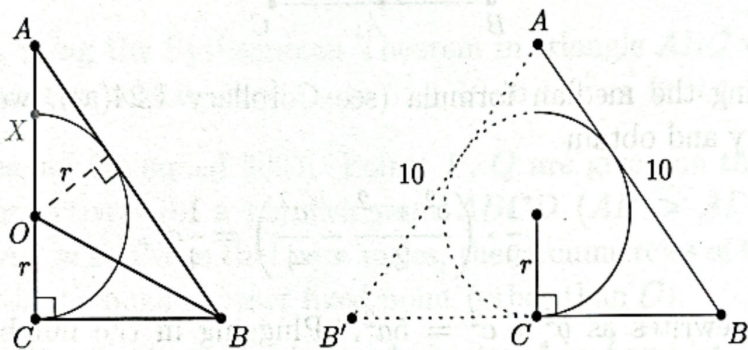
19. Let ABC be a right triangle with right angle by C and with $CA = 8$, $CB = 6$. Semicircle with diameter CX where $X \in AC$ touches side AB . Find its radius.

First Solution. Denote the center of the semicircle by O , its radius by r and the point of tangency with AB by D . We express the area of triangle ABC in two different ways.

First, as $\angle ACB = 90^\circ$, the area is simply $\frac{1}{2}AC \cdot BC = 24$. On the other hand, by the Pythagorean Theorem we have $AB = \sqrt{8^2 + 6^2} = 10$, thus

$$[ABC] = [ABO] + [BCO] = \frac{1}{2}AB \cdot r + \frac{1}{2}BC \cdot r = 8r.$$

Equating we get the answer $r = 3$.

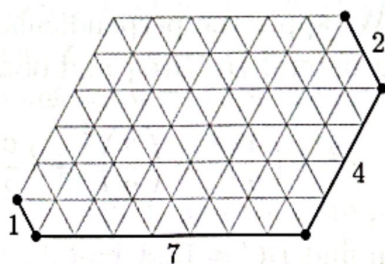


Second Solution. Denote the reflection of B about AC by B' . Then $AB' = AB = 10$, $B'B = 2 \cdot CB = 12$, and r is the inradius of triangle $AB'B$. Recalling the area formulas (see Proposition 1.25), we compute it as

$$r = \frac{[AB'B]}{\frac{1}{2}(AB' + B'B + BA)} = \frac{\frac{1}{2} \cdot 12 \cdot 8}{\frac{1}{2}(10 + 10 + 12)} = 3.$$

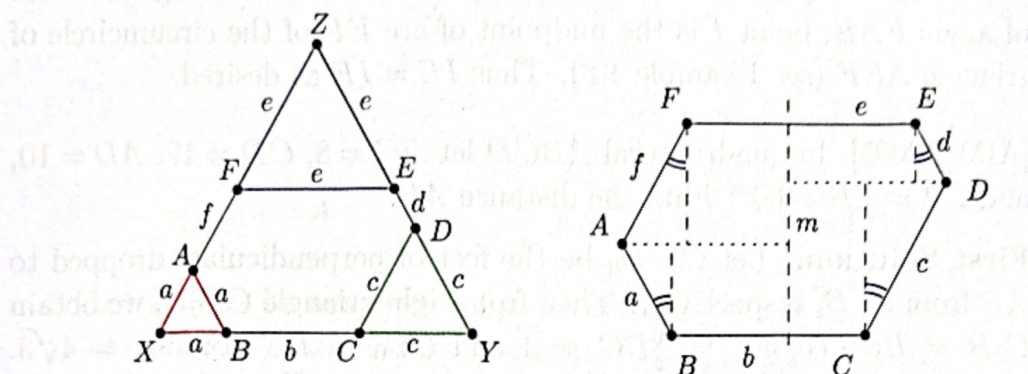
20. Four consecutive sides of an equiangular hexagon have lengths 1, 7, 4, and 2. Find the lengths of the remaining two sides.

First Solution. The common value of all the interior angles is 120° . Hence the four sides can be drawn into a triangular grid made of equilateral triangles of unit side length.



The other two sides are then seen to have lengths 6 and 5, respectively.

Second Solution. In general, let $ABCDEF$ be a hexagon with side lengths $a, b, c, d, e,$ and f , the first four of which are given, and all interior angles equal to 120° . Extending its sides AF, BC, DE we form three small equilateral triangles ABX, CDY, EFZ .



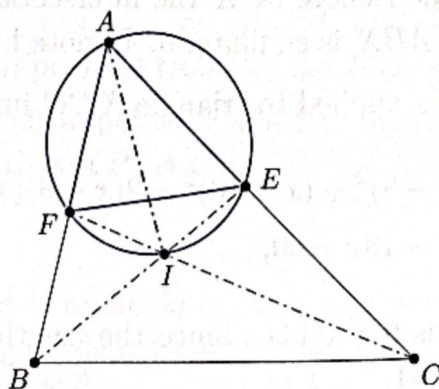
Thus triangle XYZ is equilateral too implying

$$a + b + c = c + d + e = e + f + a.$$

The side lengths e, f are then easily calculated as $e = a + b - d$ and $f = c + d - a$.

Third Solution. Place BC horizontally and observe that the sides $AB, CD, DE,$ and FA all make 30° angles with any vertical line m . Hence the lengths of the projections of these sides onto m are proportional to the side lengths. Since B and C project to the same point and so do E and F , we conclude $a + f = c + d$. Hence $f = c + d - a = 5$. Similarly rotating this argument $e = a + b - d = 6$.

21. Let ABC be a triangle with $\angle A = 60^\circ$ and denote its incenter by I . Lines BI, CI intersect the opposite sides at E, F , respectively. Prove that $IE = IF$.



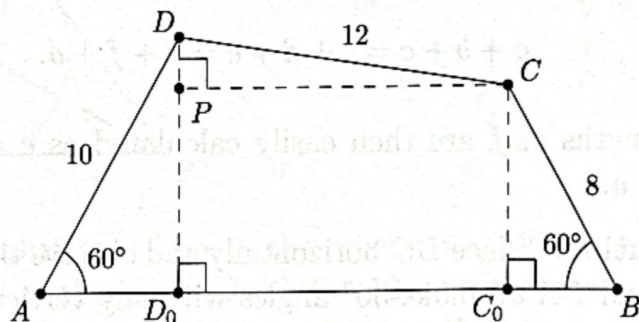
Proof. Recalling that the measure of $\angle BIC$ depends on the measure of $\angle A$ only (see Proposition 1.11) we obtain

$$\angle EIF = \angle BIC = 90^\circ + \frac{1}{2}\angle A = 120^\circ$$

and hence the quadrilateral $AFIE$ is cyclic. As AI is the angle bisector of angle FAE , point I is the midpoint of arc EF of the circumcircle of triangle AEF (see Example 1.8). Thus $IE = IF$ as desired.

22. [AIME 2005] In quadrilateral $ABCD$ let $BC = 8$, $CD = 12$, $AD = 10$, and $\angle A = \angle B = 60^\circ$. Find the distance AB .

First Solution. Let C_0, D_0 be the feet of perpendiculars dropped to AB from C, D , respectively. Then from right triangle CC_0B , we obtain $C_0B = BC \cdot \cos 60^\circ = \frac{1}{2}BC = 4$ and $CC_0 = BC \cdot \sin 60^\circ = 4\sqrt{3}$. Similarly, we get $AD_0 = \frac{1}{2}AD = 5$ and $DD_0 = 5\sqrt{3}$. Let P be the foot of perpendicular from C to DD_0 .



Then $DP = DD_0 - CC_0 = \sqrt{3}$. Rectangle D_0C_0CP gives $D_0C_0 = PC$ and from the Pythagorean Theorem in triangle DPC , we learn

$$PC = \sqrt{12^2 - (\sqrt{3})^2} = \sqrt{141}.$$

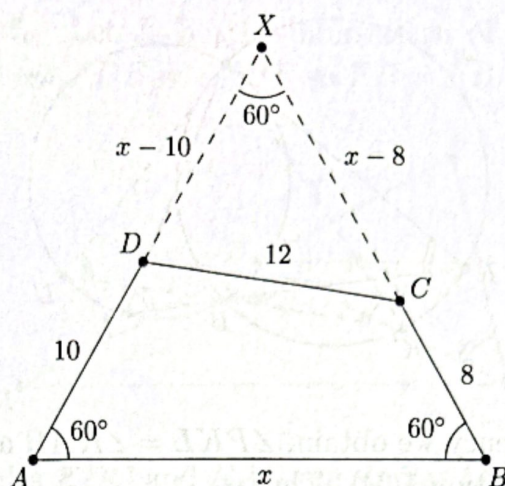
Hence $AB = AD_0 + D_0C_0 + C_0B = 9 + \sqrt{141}$.

Second Solution. Denote by X the intersection of rays AD and BC . Then the triangle ABX is equilateral. Denote by x its side length.

The Law of Cosines applied to triangle XDC implies

$$\begin{aligned} 12^2 &= (x - 8)^2 + (x - 10)^2 - 2(x - 8)(x - 10) \cos 60^\circ \\ 0 &= x^2 - 18x - 60, \end{aligned}$$

which has solutions $9 \pm \sqrt{141}$. Since the length of AX is positive, we have $AB = 9 + \sqrt{141}$.

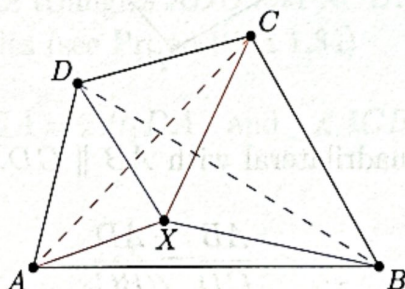


23. Let $ABCD$ be a convex quadrilateral. Find point X for which the sum of distances to its vertices is minimal.

Solution. The point we are looking for is the intersection of diagonals of $ABCD$. Indeed, by triangle inequalities in (possibly degenerate) triangles ACX , BDX we learn

$$AX + XC \geq AC \quad \text{and} \quad BX + XD \geq BD.$$

Hence $XA + XB + XC + XD \geq AC + BD$. The equality occurs if it occurs in both partial inequalities, i.e. if X lies on both AC and BD .

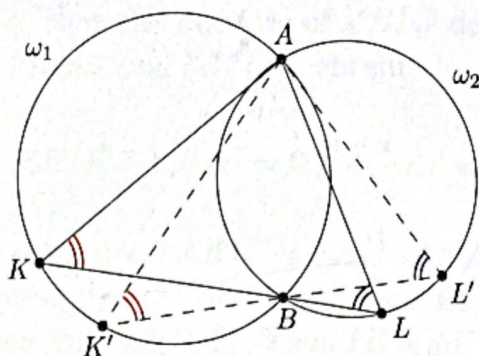


24. Circles ω_1 , ω_2 intersect at points A and B . An arbitrary line passing through B intersects ω_1 for the second time at K (outside ω_2) and ω_2 at L (outside ω_1).

- Prove that all possible triangles AKL are similar to each other.
- Let the tangents at points K and L to the respective circles intersect at P . Prove that $KPLA$ is cyclic.

Proof.

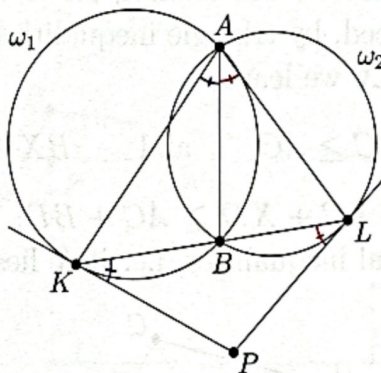
- Angle $\angle LKA$ is an angle subtending the arc AB on the circle ω_1 , hence its magnitude is fixed. Similarly, $\angle ALK$ is fixed and thus all the triangles AKL are similar (AA).



- (b) From tangency we obtain, $\angle PKL = \angle KAB$ and $\angle KLP = \angle BAL$ (see Proposition 1.34). Thus,

$$\angle LPK = 180^\circ - \angle PKL - \angle KLP = 180^\circ - \angle KAL$$

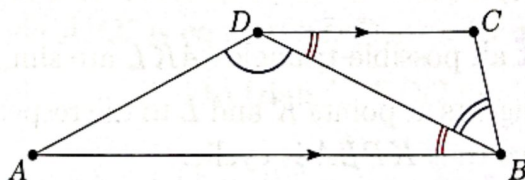
and $KPLA$ is cyclic.



25. Let $ABCD$ be a quadrilateral with $AB \parallel CD$. If $\angle ADB + \angle DBC = 180^\circ$, prove that

$$\frac{AB}{CD} = \frac{AD}{BC}.$$

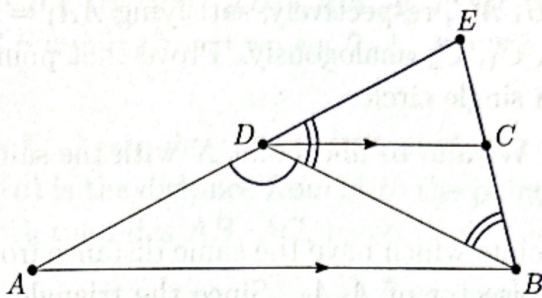
First Proof. The condition $\angle ADB + \angle DBC = 180^\circ$ suggests employing Law of Sines since $\sin \angle ADB = \sin \angle CBD$.



Observing $\angle ABD = \angle BDC$, the Law of Sines in triangles ABD and DBC implies the desired

$$\frac{AB}{AD} = \frac{\sin \angle ADB}{\sin \angle ABD} = \frac{\sin \angle CBD}{\sin \angle BDC} = \frac{CD}{BC}.$$

Second Proof. Let E be the intersection of BC and AD . Then $\angle EDB = 180^\circ - \angle ADB = \angle DBE$ so $ED = EB$.



Further, triangles EDC and EAB are similar (AA). Hence

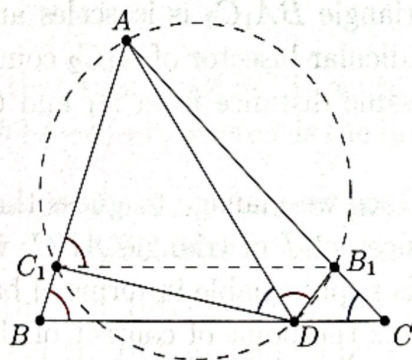
$$\frac{AB}{CD} = \frac{AE}{DE} = \frac{AE}{BE} = \frac{AD}{BC},$$

where in the last equality we again used $AB \parallel CD$.

26. [Sharygin Geometry Olympiad 2012] On side BC of triangle ABC an arbitrary point D is selected. The tangent in D to the circumcircle of triangle ABD meets AC at point B_1 . Point C_1 is defined analogously. Prove that $B_1C_1 \parallel BC$.

Proof. Keeping our diagram nice and clean we choose not to even draw the circumcircles of triangles ABD and ACD . We rewrite the tangency as equality of angles (see Proposition 1.34)

$$\angle CBA = \angle B_1DA \quad \text{and} \quad \angle ACB = \angle ADC_1.$$



Now we observe that $\angle B_1DC_1 = \angle B + \angle C$, thus it is supplementary to $\angle BAC$, which implies that the quadrilateral AC_1DB_1 is cyclic. Therefore

$$\angle B_1C_1A = \angle B_1DA = \angle B$$

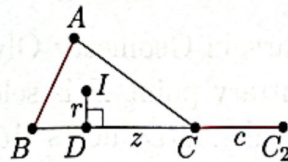
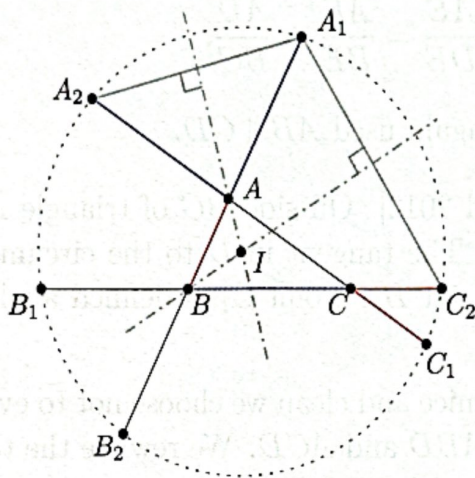
and we indeed have $B_1C_1 \parallel BC$.

27. [J. H. Conway] Conway's² circle.

Let ABC be a triangle and denote by A_1, A_2 the points on the rays opposite to AB, AC , respectively, satisfying $AA_1 = AA_2 = BC$. Define points B_1, B_2, C_1, C_2 analogously. Prove that points $A_1, A_2, B_1, B_2, C_1, C_2$ lie on a single circle.

First Proof. We aim to find point X with the same distance from all the six points.

The locus of points which have the same distance from A_1 and A_2 is the perpendicular bisector of A_1A_2 . Since the triangle AA_1A_2 is isosceles, it coincides with the angle bisector of $\angle A$. Thus the only conceivable candidate for X is the incenter I of triangle ABC .



So far we have $IA_1 = IA_2$ and likewise $IB_1 = IB_2$ and $IC_1 = IC_2$. To finish the proof, it suffices to show for instance $IA_1 = IC_2$ (the rest follows by symmetry). But this is the same thing again! As $AA_1 = BC$ and $BA = CC_2$, triangle BA_1C_2 is isosceles and the angle bisector of $\angle B$ and the perpendicular bisector of A_1C_2 coincide. Since I lies on the former, it has the same distance from A_1 and C_2 which completes the proof.

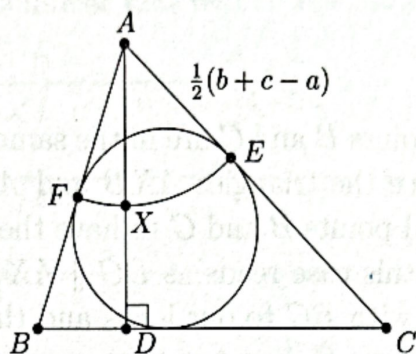
Second Proof. Once we manage to guess that the center of the circle should be the incenter I of triangle ABC , we may observe that its distance to say C_2 is approachable in terms of basic elements of triangle ABC . Indeed, if D is the point of contact of the incircle with the side BC then $DC_2 = DC + CC_2 = z + c = s$ so $IC_2 = \sqrt{r^2 + s^2}$. Since this value is symmetric in a, b, c , we may conclude.

28. Let ABC be an acute triangle. Prove that $h_a > \frac{1}{2}(b + c - a)$, where h_a is the length of A -altitude in triangle ABC .

²John Horton Conway (1936) is a contemporary British mathematician known for many delightful discoveries both in recreational and research mathematics.

First Proof. Let D be the foot of A -altitude and observe that triangle ABC being acute implies that D lies on the segment BC . Triangle inequalities in the triangles ABD and ACD yield $h_a + BD > c$ and $h_a + DC > b$. Summing them we get $2 \cdot h_a + a > b + c$ and the proof is complete.

Second Proof. Again denote the foot of A -altitude by D and recall that $\frac{1}{2}(b+c-a)$ is the distance from A to the points of contact F, E of the incircle with the sides AB, AC , respectively (see Proposition 1.15).



We want to compare the lengths AD and AE . As triangle ABC is acute, the A -altitude intersects the minor arc with center A and endpoints E, F . Denote the intersection by X . Since the whole arc EF lies inside the incircle, point X lies inside it too. Hence it lies inside the triangle ABC and thus on the segment AD . Now we are done by

$$\frac{1}{2}(b+c-a) = AE = AX < AD = h_a.$$

Remark. The conclusion is in general not true for obtuse triangles. Find an example!

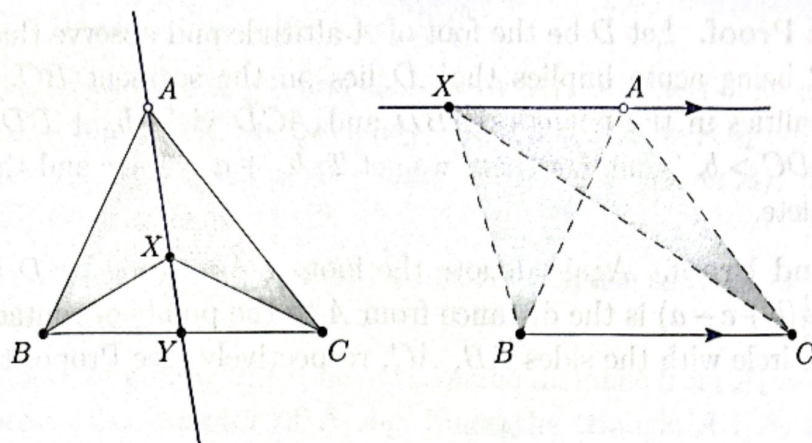
Remark. Given that triangle ABC is acute, can you prove stronger inequality $h_a > \frac{1}{2}(b+c-a) + r$, where r is the inradius of triangle ABC ?

29. Let ABC be a triangle. Find the locus of points X ($X \neq A$) for which the triangles AXB and AXC have equal area.

Solution. We distinguish two cases based on the position of X . If AX intersects the segment (!) BC , we denote by Y the intersection and use the Area Lemma (see Proposition 1.27). We have

$$\frac{[AXB]}{[AXC]} = \frac{YB}{YC},$$

so we need Y to be the midpoint of BC . In other words, in this case the desired points X form a line containing the A -median of triangle ABC .



In the other case, points B and C are in the same half-plane with borderline AX . Then since the triangles AXB and AXC have common base AX , we in fact need points B and C to have the same distance from the line AX , which in this case reads as $BC \parallel AX$. Thus, we add the line through A parallel with BC to our locus and the solution is complete.

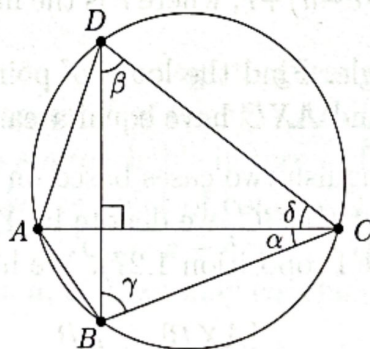
30. Let $ABCD$ be a quadrilateral with perpendicular diagonals inscribed in a circle with radius R . Prove that

$$AB^2 + BC^2 + CD^2 + DA^2 = 8R^2.$$

Proof. Denote by $\alpha, \beta, \gamma, \delta$ the inscribed angles corresponding to minor arcs AB, BC, CD, DA , respectively, of the circumcircle of $ABCD$ and rewrite each term on the left hand side by the Extended Law of Sines:

$$(2R \sin \alpha)^2 + (2R \sin \beta)^2 + (2R \sin \gamma)^2 + (2R \sin \delta)^2 = 8R^2 \quad (: 4R^2)$$

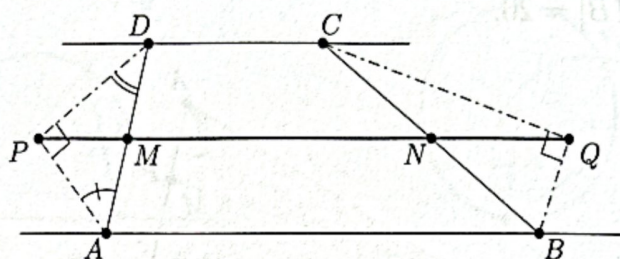
$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta = 2.$$



Since the diagonals of $ABCD$ are perpendicular we may rewrite $\sin \delta = \sin(90^\circ - \beta) = \cos \beta$ and $\sin \gamma = \sin(90^\circ - \alpha) = \cos \alpha$. The well-known identity $\sin^2(x) + \cos^2(x) = 1$ then implies the result.

31. [Mexico 1999] A trapezoid $ABCD$ has AB parallel to CD . The external bisectors of $\angle A$ and $\angle D$ meet at P , and the external bisectors of $\angle B$ and $\angle C$ meet at Q . Show that PQ is half the perimeter of $ABCD$.

First Proof. Since P lies on the external angle bisector of $\angle A$, it has equal distance from the lines AB and AD . Similarly, it has equal distance from the lines AD and CD which implies that it lies half the way between the parallel lines AB and CD . Analogous reasoning applies for Q . Thus, if we denote the midpoints of AD , BC by M , N , respectively, then P , M , N and Q are collinear and we can rewrite $PQ = PM + MN + MQ$.



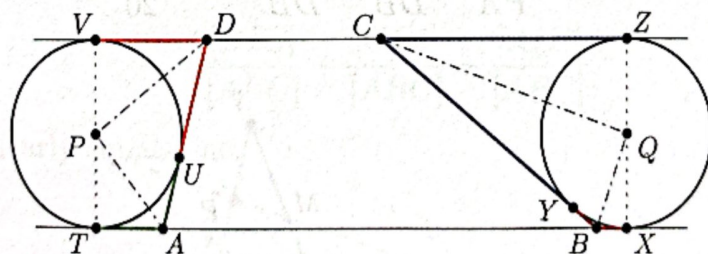
Now observe that as angles by A and D add up to 180° , the halves of their complements add up to 90° which yields $\angle APD = 90^\circ$. Hence M is the circumcenter of right triangle ADP and $MP = \frac{1}{2}AD$. For the same reason $NQ = \frac{1}{2}BC$.

Finally, since MN is the midline of trapezoid $ABCD$ we conclude by

$$PQ = PM + MN + NQ = \frac{1}{2}AD + \frac{1}{2}(AB + CD) + \frac{1}{2}BC.$$

Second Proof. Being the intersection of two angle bisectors, P is the center of a circle tangent to lines AB , AD , and CD . Denote by T , U , V the respective points of contact.

Similarly, denote by X , Y , Z the respective points of contact of the lines AB , BC , CD with corresponding circle centered at Q .



Then P , Q are the midpoints of opposite sides VT , XZ of a rectangle $VTZX$, respectively, and the result follows by Equal Tangents since

$$\begin{aligned} 2 \cdot PQ &= TX + VZ = (UA + AB + BY) + (UD + DC + CY) \\ &= AB + BC + CD + DA. \end{aligned}$$